

Lab 5 — Assignment 1: RDS PostgreSQL Deployment

Deploy and Connect Database for the Lab 4 EC2 Application (Mealie Recipe Manager)

1. Architecture

The Lab 4 EC2 instance (mealie-lab4, i-0074ba1ec324ef939) continues to run the same Mealie v3.22.0 Docker container and Caddy reverse proxy. Only the database layer changed: PostgreSQL was migrated from a local, on-instance installation to a managed Amazon RDS PostgreSQL instance in the same VPC. Client traffic flows: Client → Caddy (EC2, port 80) → Mealie container (EC2, 127.0.0.1:9000) → Amazon RDS PostgreSQL (private subnet, port 5432, not publicly reachable).

2. Database Settings

Setting	Value
Engine	PostgreSQL 16
Instance identifier	mealie-db
Instance class	db.t3.micro (Free Tier)
Storage	20 GiB (Free Tier default)
Public accessibility	No
Database name	mealie
Endpoint	mealie-db.c8b80go427hx.us-east-1.rds.amazonaws.com
Port	5432

3. Security Settings

A dedicated security group, **mealie-rds-sg**, was created specifically for RDS with a single inbound rule restricting PostgreSQL access to only the Lab 4 EC2 instance's own security group (sg-010134994e59ab4eb) — no 0.0.0.0/0 rule exists. The RDS instance itself is configured as not publicly accessible, meaning it has no route from the public internet regardless of security group rules, providing defense in depth.

Type	Port	Source
PostgreSQL	5432	EC2 security group sg-010134994e59ab4eb only

4. Data Migration

The existing Lab 4 dataset was migrated rather than starting from an empty database, using `pg_dump` and `psql`:

```
pg_dump -U mealie -h 127.0.0.1 mealie > mealie-migrate.sql
psql -h mealie-db.c8b80go427hx.us-east-1.rds.amazonaws.com -U mealie -d mealie -f
mealie-migrate.sql
```

Post-migration verification confirmed the data transferred correctly: **3 recipes** and **1 user**, matching the exact row counts present in the Lab 4 local database prior to migration.

5. Application Reconnection

Mealie's environment configuration was updated to point at the RDS endpoint and the container restarted:

```
POSTGRES_SERVER=mealie-db.c8b80go427hx.us-east-1.rds.amazonaws.com
POSTGRES_PORT=5432    POSTGRES_DB=mealie    POSTGRES_USER=mealie
$ sudo docker restart mealie
$ curl -s http://127.0.0.1:9000/api/app/about
{"production":true,"version":"v3.22.0", ...}
```

6. CRUD Demonstration

All four CRUD operations were demonstrated live through the running EC2 application, writing directly to the RDS database:

Operation	Action performed
Create	Added a new test recipe via the Mealie UI
Read	Opened an existing migrated recipe (Chicken Parmesan)
Update	Edited the test recipe's title and ingredients
Delete	Removed the test recipe

7. Evidence Summary

Screenshots captured: EC2 application running and reachable at its public IP; RDS instance showing "Available" status with the configured security group; successful pg_dump/restore output with row-count verification queries; and all four CRUD operations performed live in the browser against the RDS-backed application.